INSTITUTO UNIVERSITÁRIO DE LISBOA

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FORECASTING MODELS

BUDAPEST AIRBNB PRICE PREDICTION

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1. Introduction

As part of the Forecasting Models course, we were tasked with a project to develop a model for predicting the prices of Airbnbs in Budapest, Hungary. As requested, we focused on building models using the techniques and algorithms taught in class, namely Multivariate Linear Regression, Random Forests, Bagging and Gradient Boosting Model. We therefore avoided algorithms such as XGBoost, which are known for their effectiveness but were only mentioned in class.

1.1 Business Context

Tourism in the city of Budapest (which is the most attractive tourist destination in Hungary) has seen steady growth since the global collapse of tourism in 2020[1]. The Hungarian currency (HUF) is significantly less valuable than the Euro, and it has also depreciated considerably against the Euro over the last two decades. Even so, we should bear in mind that at the current rate, 1000 HUF is equivalent to 2.60 Euros. To make the work easier to follow, we converted the values to Euros at that rate.

Budapest is naturally divided in half by the Danube River: the left-bank side of the city is called Buda, which is the oldest part of the city, and the right-bank side is Pest, which is the more modern part with more commerce (and also where most of the tourist attractions are located).

2. DATA UNDERSTANDING

The Data Understanding phase begins with the initial collection of the data and continues with activities aimed at becoming familiar with them, identifying quality problems, discovering preliminary *insights* and detecting interesting subsets for forming hypotheses.

2.1 Data Description

A preliminary analysis of the data structure was carried out to classify the available variables and their predictive potential. The dataset consists of the following variables:

- **price:** Continuous numerical variable representing the daily cost of the accommodation, which is the target variable of our work.
- **neighbourhood:** Categorical variable representing the districts of Budapest in Roman numerals.
- **latitude** and **longitude:** Numerical variables representing the exact spatial coordinates of the accommodation.
- **room_type:** Categorical variable representing the type of accommodation (Private or shared room, apartment, etc…)
- **minimum_nights:** Discrete numerical variable representing the minimum number of rental nights accepted.
- **availability_365:** Discrete numerical variable representing the number of days on which the accommodation will be available over the next 365 days.

- **number_of_reviews, reviews_per_month and number_of_reviews_ltm:** Numerical variables representing, respectively, the total number of reviews of the accommodation, the average number of reviews per month and the number of reviews in the last twelve months
- **last_review:** Numerical variable indicating the date of the last review of the accommodation in question.

Created variables:

- **days_since_last_review:** Numerical variable representing the number of days since the last review, based on last_review
- **dist_centro:** Numerical variable representing a pseudo-Euclidean distance between the coordinates of an accommodation and the “centre” of the city's tourist activity.

Metadata (to be excluded from modelling):

- Unique identifiers (id, host_id) and unstructured text descriptions (name, host_name, license), plus a column of blank values (neighbourhood_group).

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[1] *Number of tourist arrivals (nearly 1.7 million) up 11% year-on-year and tourism nights (3.8 million) up 10%* - https://www.ksh.hu/en/first-releases/tsz/etsz2510.html

2.2 Data Quality Verification

Before any in-depth exploratory analysis, the “health” of the data was assessed in order to detect anomalies that could compromise the modelling; this turned out to be the most difficult part of the work.

When we analysed the database in general terms (removing the rows where price was empty, since they are of no use for our work) using the summary function, or even by investigating it manually, we found several odd cases, some of them systematic:

```
> summary(dados)
       id                name             host_id          host_name              neighbourhood     latitude       longitude              room_type         price        
 Length:11577       Length:11577       Length:11577       Length:11577       VII. kerület :3335   Min.   :47.39   Min.   :18.94   Entire home/apt:10746   Min.   :   10.18  
 Class :character   Class :character   Class :character   Class :character   VI. kerület  :2020   1st Qu.:47.49   1st Qu.:19.06   Hotel room     :    6   1st Qu.:   41.65  
 Mode  :character   Mode  :character   Mode  :character   Mode  :character   VIII. kerület:1736   Median :47.50   Median :19.06   Private room   :  784   Median :   56.91  
                                                                            V. kerület   :1280   Mean   :47.50   Mean   :19.06   Shared room    :   41   Mean   :  100.25  
                                                                            IX. kerület  : 841   3rd Qu.:47.51   3rd Qu.:19.07                           3rd Qu.:   85.08  
                                                                            XIII. kerület: 758   Max.   :47.61   Max.   :19.32                           Max.   :81837.15  
                                                                            (Other)      :1607                                                                     
 minimum_nights    number_of_reviews  last_review         reviews_per_month  calculated_host_listings_count availability_365 number_of_reviews_ltm   license         
 Min.   :  1.000   Min.   :   0.00   Min.   :2012-05-05   Min.   : 0.010   Min.   :  1.00                 Min.   :  0.0    Min.   :  0.00        Length:11577      
 1st Qu.:  1.000   1st Qu.:   8.00   1st Qu.:2025-08-04   1st Qu.: 0.730   1st Qu.:  1.00                 1st Qu.: 79.0    1st Qu.:  2.00        Class :character  
 Median :  2.000   Median :  39.00   Median :2025-09-04   Median : 1.920   Median :  4.00                 Median :189.0    Median : 13.00        Mode  :character  
 Mean   :  3.273   Mean   :  88.51   Mean   :2025-06-07   Mean   : 2.288   Mean   : 46.93                 Mean   :189.4    Mean   : 21.03                          
 3rd Qu.:  2.000   3rd Qu.: 118.00   3rd Qu.:2025-09-14   3rd Qu.: 3.520   3rd Qu.: 29.00                 3rd Qu.:299.0    3rd Qu.: 35.00                          
 Max.   :730.000   Max.   :1180.00   Max.   :2025-09-24   Max.   :21.670   Max.   :396.00                 Max.   :365.0    Max.   :199.00                          
                                     NA's   :868          NA's   :868                                                                                             
 days_since_last_review
 Min.   :   0.0        
 1st Qu.:  10.0        
 Median :  20.0        
 Mean   : 108.6        
 3rd Qu.:  51.0        
 Max.   :4890.0        
 NA's   :868           
```

The first two aspects to note are the maximum rental cost for one night compared with the third quartile. Even though 81837 euros is a cost per night that is easily verifiable as an error (if only because of its 0 reviews), there are quite a few other values behind it that are very surprising. Another value that seems to be wrong is minimum_nights, which has a third quartile of 2 nights but a maximum of 730 nights. To understand the data even better, we filtered the database into smaller ones using

specific conditions, for example long stays (minimum_nights > 30) or no reviews (number_of_reviews == 0).

The main obstacles to detecting outliers are that we are aware of the existence of luxury accommodation, but have no way of recognising it, since the data we have on each listing are quite limited. In addition, it seems that some of the observations with a minimum number of nights greater than 30 are using the price variable to represent not the price per night but the price per month. One of the salient examples of these inconsistencies is ID = 1283159: it is hard to believe that a house costing 1017 euros per night, with a minimum stay of 80 nights, actually has 432 reviews, especially since, even assuming that 432 is the total number of clients (a family counts as one client in the reviews) who stayed there, with a minimum of 80 nights each, this would imply that the house has been fully booked for at least 94 years! This, together with the fact that the last review of this accommodation is more than three years old, seems sufficient indication that minimum_nights alone, or together with the price, was deliberately increased to keep new clients away without having to delete the listing while the accommodation is unavailable.

One way of trying to deal with this type of “dead” listing would be to filter the data by the time since the last review, which is why we created the variable days_since_last_review; but in doing so we solved one problem by adding a new one, since out of a total of almost 12000 observations, 900 have no reviews. We could assume that listings without reviews are explained by their being very new, but we find it at least as plausible that they are listings with values that are too high for what they offer and are therefore “rejected” by the market.

Unable to agree on a way of defining outliers, we ended up simply using the interquartile range (i.e., 1.5 IQR above Q3 and below Q1) to remove upper and lower values from the database. This radical cut is generally considered bad practice, since in this way we are training a model to predict only part of the real values (albeit a large majority of them). We also opted for it because we recognised that, with the data we had on each listing, it would be very difficult to build a model with any kind of relevance if we did not do so. In this way, we partly rid ourselves of the difficulty of working out what luxury accommodation is, since the ceiling created by the calculation mentioned ended up at around 150 Euros.

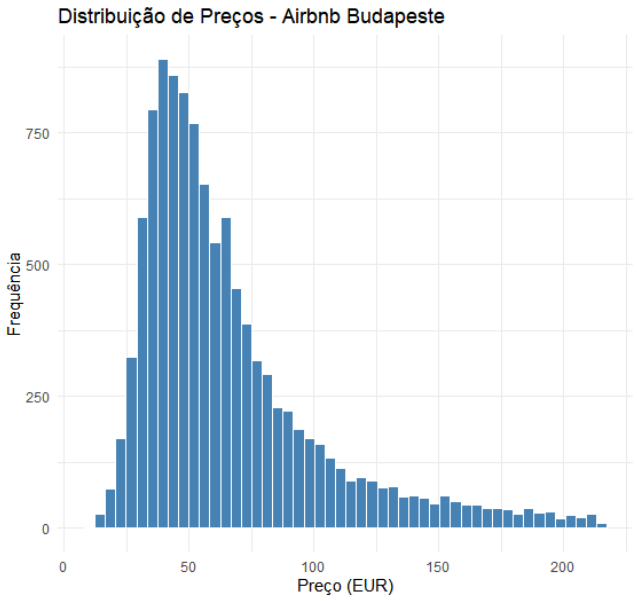
2.3 Exploratory Data Analysis (EDA)

With the database now clean, we began to analyse what remained.

2.3.1 Exploratory Analysis: Price Distribution

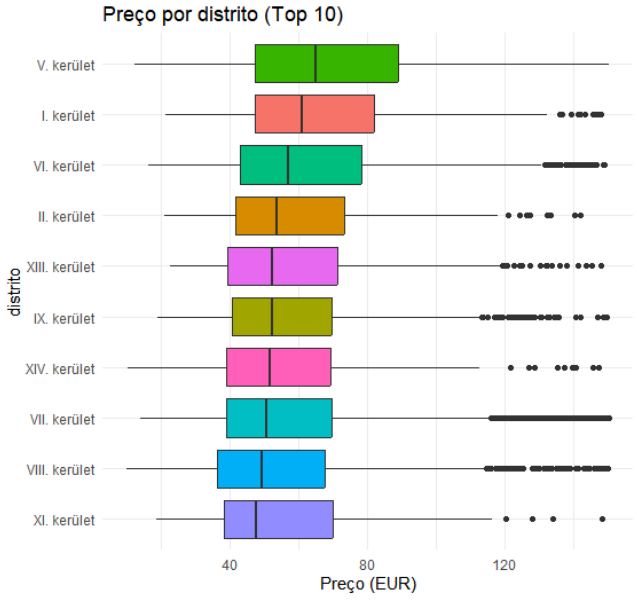
The histogram of the price variable shows a distribution with strong positive skewness. The vast majority of observations are concentrated in an affordable price range, with a long tail extending towards high values.

- This non-normal distribution suggests that the RMSE evaluation metric may be disproportionately affected by high values even after cleaning the database.
- The long tail towards higher values is common in this type of data, as is the *skewness* that makes lower values much more frequently observed.



The graphical analysis "Price by District (Top 10)" reveals some regional disparities.

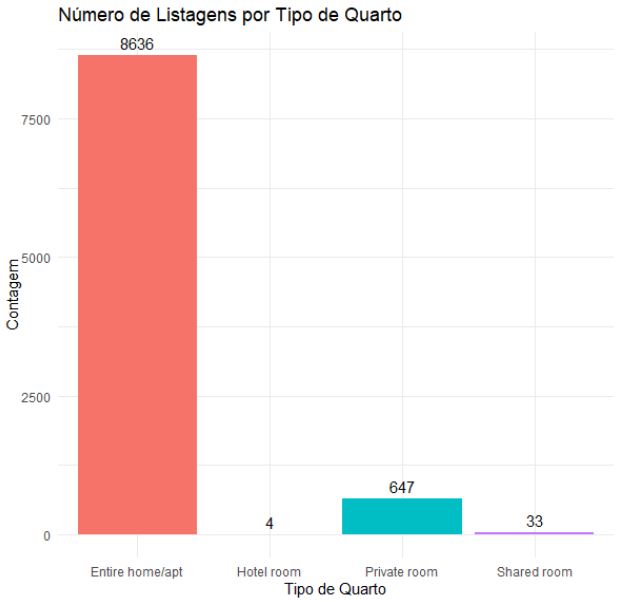
- District V stands out with the highest median price, which is consistent with what would be expected (it is the tourist centre, where the Parliament and the Basilica are located).
- In general the values do not seem to diverge much; this may be due to the overly radical cut we made, but looking at the order of the medians, it follows what would be expected when looking at *Annex 1*.



In *Annex 2* we can also see a chart with the spatial distribution of the listings and how the density of accommodation is much higher in Pest than in Buda.

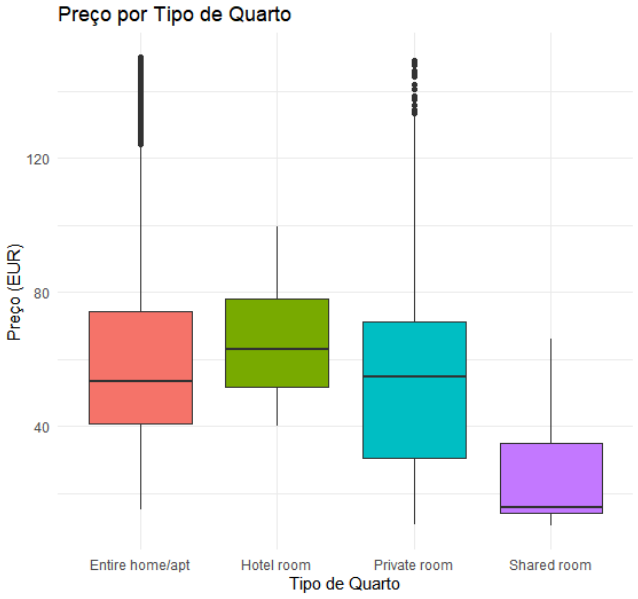
2.3.3 Exploratory Analysis: Accommodation Type (room_type)

1. *Entire home/apt* is clearly the majority within our dataset, with *private room* still being significant.



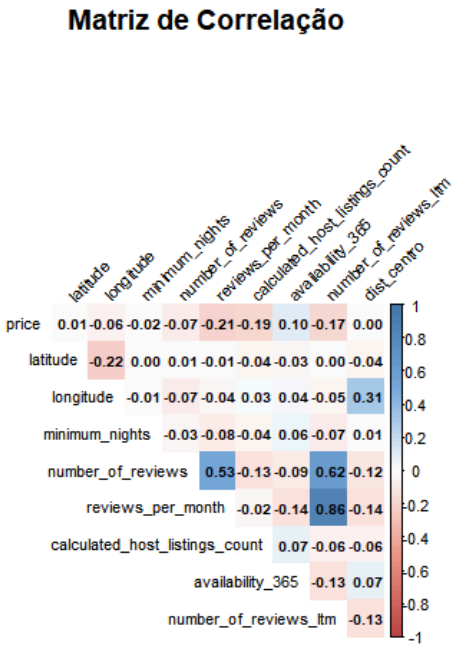
The *Boxplot* chart confirms what would be expected:

1. Shared room has the lowest values
2. Private room varies considerably in price, even more than apartments, but both have similar values.
3. Hotel rooms have a much smaller dispersion than expected. This is due to the small number of listings they have in the dataset.



2.3.4 Correlation Matrix

The correlation analysis (Pearson) between numerical variables indicated that there is some correlation among the variables relating to the number of reviews, but, above all, a strong positive correlation between reviews_per_month and number_of_review_ltm. A value above 0.80 is generally a sign that one of the variables should be removed from the model, but in this case we feel that reviews_per_month carries important information that is different from the total number of reviews in the last 12 months. The value itself is a surprise, as we would have expected number_of_reviews to have a higher correlation with reviews_per_month.



3. DATA PREPARATION

3.1 Data Selection

The first task consisted of filtering the original *dataset* to keep only the columns relevant to the regression problem, excluding noise and redundancy.

- **Excluded Attributes:**
  - **Identifiers (id, host_id):** Variables with unique or very high cardinality that have no generalisable pattern.
  - **Free Text (name, host_name, license):** Although they contain potential information, including them would require *Text Mining* techniques, which we felt were not the aim of the work.
  - **Created variable:** days_since_last_review, since we found no way of handling listings without any reviews (we assume that if we assigned them some arbitrary very high number, we would be biasing the models with information that is not real)
- **Selected Attributes:**
  - The variables identified in the Exploratory Analysis as having predictive potential were kept: neighbourhood, latitude, longitude, room_type, minimum_nights, number_of_reviews, reviews_per_month, number_of_reviews_ltm, availability_365, dist_centro and the target variable price.

3.2 Handling Missing Values (NAs)

The NAs of the number_of_reviews variables were replaced by 0 in the initial stages of exploring the database. The same was done for the few observations with NA for

the price variable.

4. MODELING

We used the *caret* package to train the models and an 80% Training / 20% Test split for the sets.

4.1 Testing and Validation Strategy

Before running the algorithms, we defined the validation strategy to ensure that the results do not suffer from *overfitting*. We therefore decided to use cross-validation, ensuring that the model is exposed to different variations of the training data before the final evaluation. We considered *10-fold* sufficient for this purpose.

4.2 Model 1: Multiple Linear Regression

The first model trained was Multiple Linear Regression.

Given the work carried out up to this point, it was expected to be the model that would suffer most from the lack of explanatory variables and the expected non-linearity of the relationships between price and the remaining variables. Its ability to explain the impact of each variable was also affected by the use of highly correlated variables.

4.3 Model 2: Bagging

Since we were supposed to train models based on decision trees, but with algorithms that improve on them, we did not train any individual decision tree. The first model of this type to be trained was Bagging.

The algorithm generates different training datasets through sampling with replacement. For each sample, it trains a full decision tree (without pruning). The final prediction is the average of the predictions of all the trees. We chose to use 100 trees (nbagg = 100).

4.4 Model 3: Random Forest

Whereas Bagging considers all variables to find the best "split" at each node of the tree, Random Forest considers only a random subset of variables (mtry) at each split.

The main advantage of the model is that, by forcing the trees to use different variables, a very strong variable is prevented from dominating all the trees, allowing more subtle variables to contribute to the model.

The model was trained with 100 trees (ntree = 100) and, instead of forcing a specific mtry, we used an mtry_grid, with the mtry values to be tested being (1, 2, 3, 4, 6, 8, 10).

4.5 Model 4: Gradient Boosting Machines (GBM)

Finally, Gradient Boosting was applied, a *Boosting* technique rather than *Bagging*.

Unlike Random Forest, which builds independent trees in parallel, GBM builds trees sequentially. Each new tree is trained to predict the residuals of the previous tree,

progressively correcting the model.

We used tunelength=3 in this model.

5. EVALUATION

After the models had been trained, we created a table to compare their training results using three metrics: RMSE, MAE and R^2

5.1 Comparative Analysis of Metrics

- **MAE (*Mean Absolute Error*):** Measures the mean absolute error, and is easier to interpret financially.
- **RMSE (*Root Mean Squared Error*):** Penalises large errors. Since we made a large upper cut in accommodation values, this metric will be more important than MAE.
- **R^2 (*Coefficient of Determination*):** Indicates the percentage of the variance in price that is explained by the model. Generally considered the main metric in modelling.

5.2 Training Results

Linear Regression

```
> print(modelo_lm)
Linear Regression

8365 samples
  11 predictor

No pre-processing
Resampling: Cross-Validated (10 fold)
Summary of sample sizes: 7529, 7528, 7529, 7527, 7529, 7529, ...
Resampling results:

  RMSE      Rsquared   MAE
  25.21739  0.1338433  19.50919
```

Bagging

```
Bagged CART

8365 samples
  11 predictor

No pre-processing
Resampling: Cross-Validated (10 fold)
Summary of sample sizes: 7528, 7528, 7528, 7529, 7529, 7528, ...
Resampling results:

  RMSE      Rsquared   MAE
  25.15685  0.1413824  19.29525
```

Random Forest

```
  mtry  RMSE      Rsquared   MAE
   1    26.10377  0.1551180  20.37658
   2    25.04711  0.1929570  19.35890
   3    24.31331  0.2244769  18.63422
   4    23.82785  0.2442981  18.11275
   6    23.29770  0.2678338  17.54812
   8    23.12792  0.2744768  17.31838
  10    23.07563  0.2767526  17.23100

RMSE was used to select the optimal model using the smallest value.
The final value used for the model was mtry = 10.
```

GBM

```
  interaction.depth  n.trees  RMSE      Rsquared   MAE
  1                   50      25.27539  0.1407902  19.49381
  1                  100      24.89858  0.1610780  19.14401
  1                  150      24.71968  0.1697996  18.94390
  2                   50      24.73428  0.1765598  18.97468
  2                  100      24.35770  0.1946155  18.58835
  2                  150      24.19370  0.2036207  18.40916
  3                   50      24.42054  0.1953399  18.68506
  3                  100      24.09317  0.2111996  18.31412
  3                  150      23.94797  0.2193473  18.16555

Tuning parameter 'shrinkage' was held constant at a value of 0.1
Tuning parameter 'n.minobsinnode' was held constant at a value of 10
RMSE was used to select the optimal model using the smallest value.
The final values used for the model were n.trees = 150, interaction.depth = 3, shrinkage = 0.1 and n.minobsinnode = 10.
>
```

Analysis of the results:

1. **Linear Regression and Bagging:** Showed the worst performance (highest RMSE and lowest R^2). Both surprisingly close.
2. **Gradient Boosting:** Showed a substantial improvement over linear regression and bagging, especially with regard to R^2.
3. **Random Forest:** Best on all metrics; it outperformed Gradient Boosting mainly in R^2.

We therefore crowned the Random Forest model with mtry=10 as the model for making the predictions on the Test set:

6. RESULTS

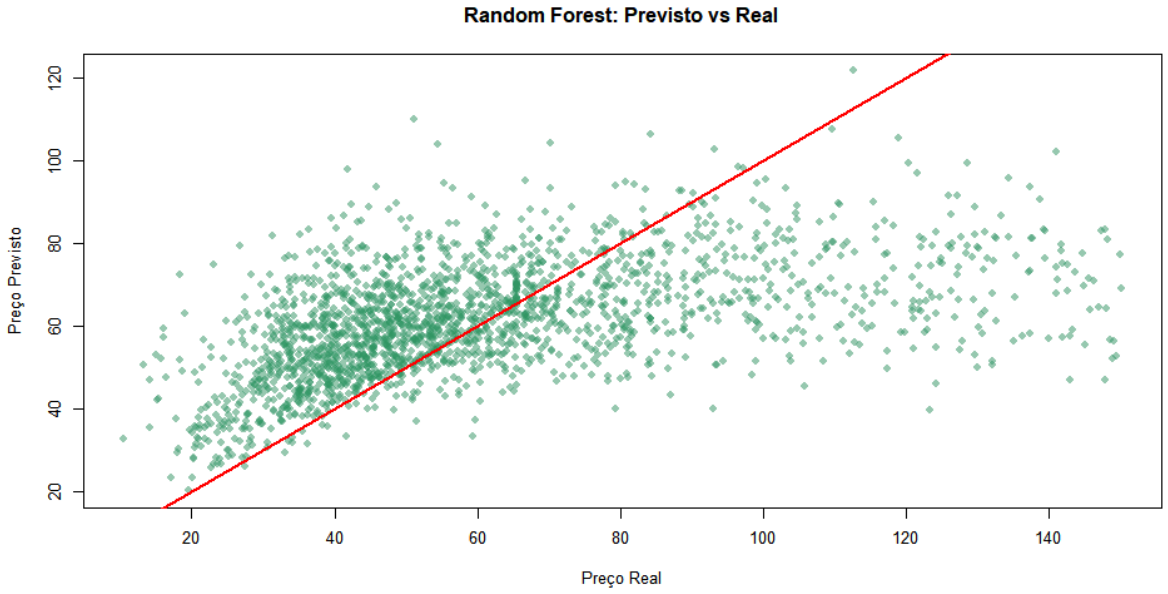
After testing, we obtained the following results:

Model	RMSE	MAE	R2
Random Forest	24.17	17.79	0.2627

We can see that the values are quite close to the training results and that the cross-validation succeeded in avoiding *overfitting*.

6.1 Graphical Analysis: Predicted vs. Actual

Visual analysis of the scatter plot:

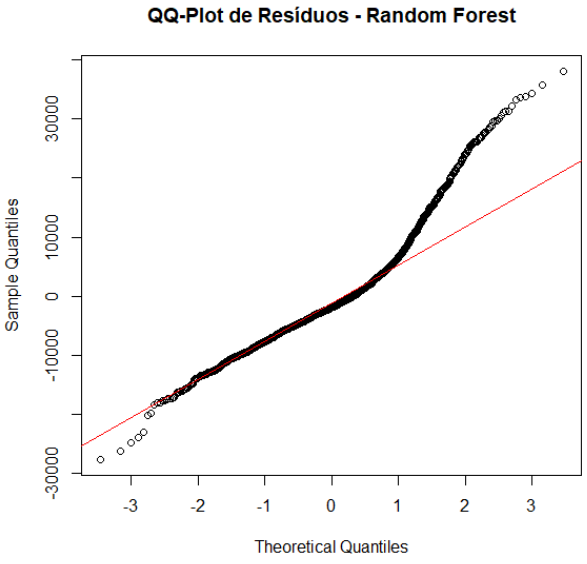


The scatter plot confirms quite well our idea that the model (whichever turned out to be the best) would have great difficulty predicting large values, because the database did not contain the variables that best explain price variations; for example, the difference between a private room at 50 euros per night or 150 may be the installation of air conditioning or the number of beds. From 70 actual euros upwards, our model seems to refuse to predict anything above 80 euros per night. Even so, it manages to follow the increase in value up to that point to some extent. Between 20 and 60 it appears to prefer overvaluing the accommodation, but the “patch” of overvaluation seems to follow, even if not very faithfully, the perfect prediction line.

6.2 Residual Analysis and Statistical Validity

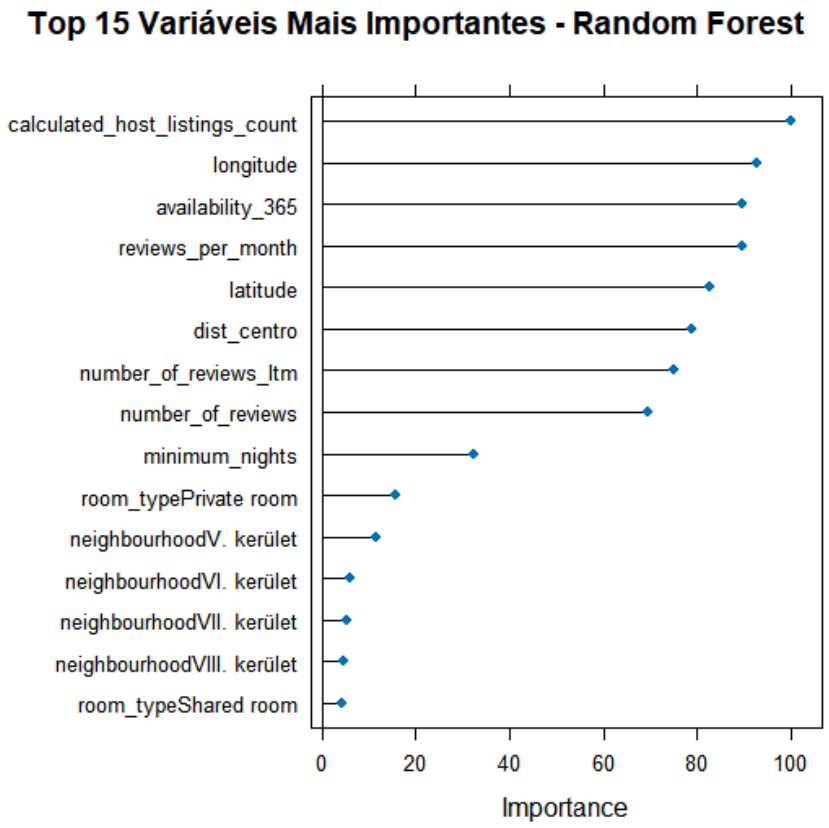
To validate the robustness of the model, the distribution of the errors (residuals) was analysed.

The residual analysis via QQ-Plot shows that the Random Forest model exhibits robust and statistically valid behaviour for the standard segment (which represents the majority of the listings), as evidenced by the normal distribution of the residuals in the central range. However, significant deviations are observed in the tails of the distribution, indicating systematic errors in the prediction of extreme values. This limitation is probably due to the scarcity of variables that explain price increases and, at the opposite extreme, what makes an accommodation so cheap.



6.4 Model Interpretability / Variable Importance

Analysis of variable importance is essential for turning the “black-box” nature of Random Forest into something interpretable, making it possible to identify which predictors contribute most to price variation and to validate the theoretical coherence of the data analysed:



The hierarchy of variable importance was a surprise. The city district, or even the room type, which one would expect to be among the most important variables, do not even make it into the top 10. But there are ways of interpreting several of these values:

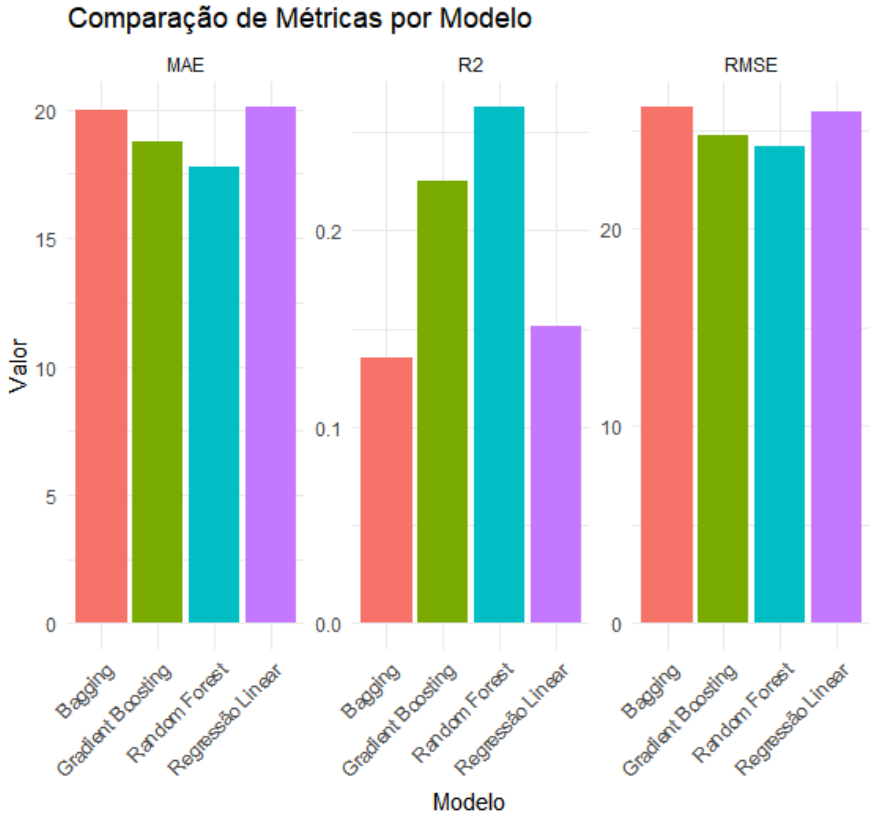
Possibly, the “irrelevance” of the districts arises simply from the introduction of the variable dist_centro, whose coordinates are somewhere near the centre of the busy area of Pest, the area known for its commerce. A redundancy is created, since the distance of the accommodation from the centre is directly related to the District in which it is located. The same can be said of the two even more obvious variables “latitude” and “longitude”. The three together mean that the districts by themselves are not important in the trees' decisions.

It makes sense that the review-related variables are important, since they are a sign that the owner of the accommodation charges a fair price for what they sell, which makes the model's prediction task easier. Similar reasoning should be applied to the most important variable. Possibly, a host with more experience in the area (with more accommodations) will tend to know the market better and to apply more justifiable, and therefore more predictable, prices.

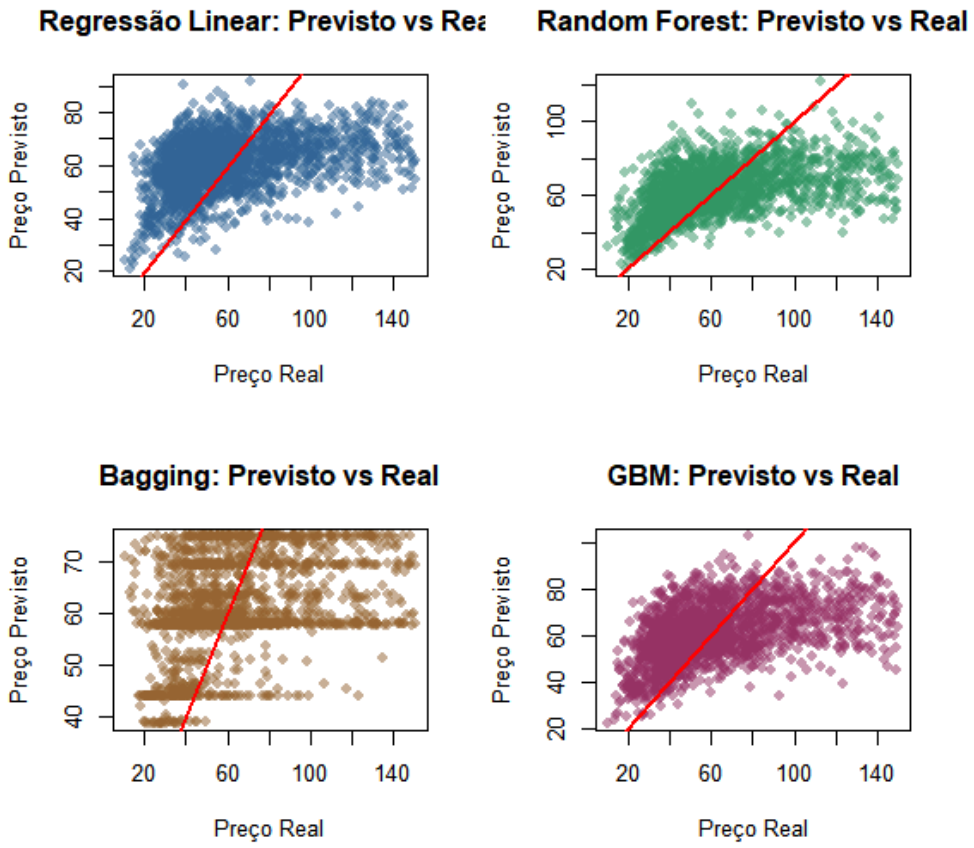
The importance of minimum_nights, even though smaller, may reflect the distinction between two different markets: the short-stay tourism market (stays of 1 to 3 nights, with higher prices per night) and the medium/long-stay or digital-nomad market (stays > 30 nights, with lower base prices). The model uses this variable as a filter to apply different pricing logics to these two distinct segments.

6.5 Comparison

Even though the aim of the work was to use only the best model to make the final prediction, we ended up testing all of them to obtain a graphical comparison of their performance:



The metrics for all the models corroborate the training results, with the exception of the R^2 of the Linear Regression model, which seems to have increased slightly, unexpectedly leaving Bagging as the model with the worst performance in both RMSE and R^2.



Linear regression performs much closer to Random Forest and GBM than expected, but it is easy to recognise that the “cloud” it creates is considerably less centred on the perfect prediction line. It is also important to note the scales of the predicted prices, since these depend on each model's maximum prediction.

Bagging is the model with the least natural prediction chart. It appears to be suffering severely from *underfitting*. We can observe a stratification of prices that should not be seen in a bagging model, in addition to a completely defined maximum ceiling.

7. CONCLUSION

The prediction results, especially R^2, for the final model are close to, or even better than, what would be expected given the limitations of the database, but this will be related to the way the outliers were identified (through the IQR, which is quite generic and does not represent well the price disparity between the overwhelming majority of the market, which is quite affordable, and the luxury stays that reach values far above what would be expected, which are much less representative of the market, while still being something that good practice says should not be ignored.

The absence of data such as the inclusion of air conditioning, the number of bedrooms and bathrooms or the area of the accommodation creates a barrier to the models' prediction of the more expensive accommodation. We verified exactly this with the cut we made at around 150 Euros per night.

Acknowledging the limitation of our conclusions given the outlier cut applied to the database, we believe that we have produced a piece of work with methodological honesty.